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Electrophysiological responses to digit stimulation in a tactile oddball paradigm

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ABSTRACT

Sensory memory traces are assessed via oddball paradigms in which deviant (infrequent) stimuli are interspersed into a string of standard (frequent) stimuli. Once a memory trace for the standard is established, the deviant spurs a change detection response measured via the resulting event related potential (ERP). Response magnitude is sensitive to the differences in stimuli properties or categories and influenced by individual experience. The goal of the present study was to use ERPs to test the relation between individual digits in the somatosensory cortex and the extent to which digit representations are influenced by individual differences in experience such as independent mobility and playing video games. The present study of 60 undergraduates utilized a passive tactile oddball paradigm, stimulating the thumb, middle, and little fingers. The oddball paradigm was fully matched with each digit serving as the standard and deviant. A temporal principal component analysis (tPCA) identified factors that matched three a priori ERP components: N80, somatosensory mismatch negativity (sMMN), and P300. Analyses confirmed the anticipated differences between standards and deviants and provided some support for prior ERP work suggesting the thumb is in a different functional category than the other digits. Independent control of individual digits (such as the little finger) was positively related to only one aspect of the ERP (P3a) while video game experience was not associated with ERP differences. Cumulatively, these results provide a more nuanced examination of tactile oddball paradigms and how ERP methods can shed light on the relations between different digits.

1. Introduction

Within cognitive neuroscience, research on neural responses to changes in a train of repetitive sensory stimuli has shed light on multiple aspects of attention (Näätänen et al., 2019). One central construct in this area is sensory memory, demonstrated via the detection and prediction of changes or deviations in sensory stimulation. A commonly used method to probe sensory memory utilizes oddball paradigms that combine repetitive frequent (standard) stimuli with infrequent (oddball) stimuli that differ in some physical characteristic. Change detection is quantified using event-related potentials (ERP), recorded via electroencephalogram (EEG) (Näätänen et al., 2019).

When the sensory memory trace to a standard (frequent) stimulus has been formed, a sudden interruption by the deviant (infrequent) stimulus produces a change detection response in the ERP (Friedman et al., 2001; Fu and Monahan, 2021). The magnitude of these responses is dependent on the salience of the discrepancy between the standard

and deviant stimuli and changes in the ERP occur at multiple times following stimulus onset (Calcus et al., 2015; Näätänen and Alho, 1997).

Between 100 and 250 ms, the mismatch negativity (MMN), a negativity in the difference waveform between the standard and deviant stimuli is visible (Duncan et al., 2009; Tsolaki et al., 2015). The magnitude of the MMN response is sensitive to the extent of difference between the standard and the deviant (Duncan et al., 2009; Näätänen and Alho, 1997) which is, in turn, influenced by participants' prior experience with the stimulus contrasts (Genç et al., 2009). For example, participants with extensive piano training display larger auditory MMN responses than those without musical training (Olszewska et al., 2021). Later in the waveform, the 'novelty P300' or P3a, a positive peak over midline electrodes is visible (Barry et al., 2020; Polich, 2007). Greater differences between the standard and deviant produce larger P3a amplitudes and shorter latencies (Duncan et al., 2009; Tsolaki et al., 2015; Zheng et al., 2012).

Sensory memory research in the auditory domain has varied either

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physical properties of the stimuli (e.g., tone or volume) or the stimulus category (e.g., similar pitch contours) (Tsolaki et al., 2015; Zheng et al., 2012, respectively). The research on categorization is driven by the notion that differences in the elicited ERPs will be greater when the standard and deviant stimuli cross a categorical boundary than when the stimuli are perceived as being within the same functional category (Shen and Froud, 2019). Emerging research has tested whether this categorical principal generalizes to other sensory domains to understand whether, for example, brain responses during tactile oddball paradigms can be used to probe the segmentation of the body into different functional parts (Shen et al., 2018a, 2018b, 2020). In the current study, we utilized a passive oddball paradigm involving tactile stimulation that was designed to test the relations between different fingers of the hand. Our working hypothesis was that variation in patterns of neural responses to stimulation of different digit pairs would relate to individual differences in the ability to control fingers independently.

1.1. Tactile oddball paradigms

Oddball paradigms in the tactile modality have the potential to shed light on the neuroplasticity of the somatosensory cortex, but broad conclusions are stifled by methodological variability. Experiments vary in methods of stimulus delivery, spanning electrical and mechanical (e.g., vibrotactile, applied force) techniques, and deviant stimuli characteristics which may differ in intensity, frequency, or location compared with the standard stimulus (Näätänen et al., 2019). Results across paradigms are consistent in eliciting and registering a change in the somatosensory evoked potential (SEP) to deviant stimuli, but there is variability in how that change is manifested (Akatsuka et al., 2005, 2007; Shen et al., 2018a, 2018b). For example, Akatsuka et al. (2005) presented a train of electrical stimuli to the back of the hand, with the deviant condition consisting of a change in the duration between tactile stimuli while another study used electrical stimuli across different locations (different fingers) (Strömmer et al., 2014). These research groups found different SEP components, summarized in Table 1, which differ further still from components evoked in auditory paradigms. Interestingly, two extant studies using mechanical stimulation (e.g., a tap from an external object) on different body locations found the MMN and P3a components in latency ranges that were comparable to findings in the auditory modality (Shen et al., 2018b, 2018a).

The discrepancies in SEP responses to different oddball paradigms may be related to differences in the stimuli, task protocols, or in the derivation of SEP components. Historically, the main method of ERP scoring includes visual inspection of grand mean waveforms at the individual or group level to identify time windows for ERP components by selecting windows around either positive or negative peaks in the data. Within these windows, researchers may use multiple scoring methods

such as extracting the peak or mean amplitude (Luck, 2014). Studies of SEP responses in tactile oddball paradigms have differed in scoring methods and time windows of interest (see Table 1). Partly in response to these inconsistencies, there has been a push to move away from visual scoring and that relatively arbitrary selection of time windows leads to ERP components that are less than robust (Luck and Gaspelin, 2017). In comparison to the visual scoring method, data-driven approaches such as a temporal principal component analysis (tPCA), which identifies unique factors within the ERP response that have their own topography, latencies, and waveforms (Scharf et al., 2022). Researchers then assess these factors for their fit with the hypothesized ERP components of interest.

The use of PCA has been encouraged for the discrimination of sub-components of P300 such as the P3a and P3b (Polich, 2007), but this method has rarely been run to discern the MMN, particularly with tactile stimuli. We suggest that using the PCA holds promise for the study of brain responses derived from oddball paradigms, including the tactile MMN, and we aimed to apply this approach in the current study.

1.2. Research aims

Study aims were centered around replicating and extending the tactile oddball paradigm utilized by Shen et al. (2018a) that employed mechanical stimulation of different digits of the hand. In that study, three digits of the right hand were stimulated: The middle (3rd) digit served as the location of standard stimuli, with deviant stimuli being presented to the thumb and little fingers (1st and 5th digits). One main finding of Shen et al. (2018a) was that the amplitude of the sMMN elicited by a contrast of the third digit and the first digit (the thumb) was significantly larger than the sMMN elicited by the contrast between the third and fifth digits. The authors suggested that this finding reflects a functional boundary effect that is related to the way that objects are typically grasped. They also suggested that this boundary effect could be further investigated by studying populations in which the functional role of each individual finger may be more distinct, for instance in musicians or video game players.

A limitation of the study of Shen et al. (2018a) was that only one site of standard stimulation was used, which was the third digit. In the current study, we expanded upon the design of Shen et al. (2018a) and ran additional oddball blocks in which each of the three digits (1, 3, and 5) was the site of standard stimulation. Through using each digit as the standard, we extended the findings of Shen et al. (2018a) by examining sMMN and P3a responses to a wider range of digit combinations. In the current study we were also interested in how individual differences in finger dexterity and video game experience relate to ERP responses elicited using a tactile oddball paradigm. Studies of musicians (pianists and string players) have shown that variation in brain responses to

Table 1
SEP scoring criteria across studies using tactile oddball paradigms. Shading is matched across components with darker hues representing later components.

Article	Component	Peak/Average	Time window
(Akatsuka et al., 2005)	N60	Peak and average (all window)	40-70ms
	P150	Peak and average (all window)	100-200ms
(Strömmer et al., 2014)	N80	Peak	40-110ms
	Early MMR	Peak	180-220ms
	Late MMR	Peak	250-290ms
(Shen et al., 2018a)	N80	Average (12 ms window)	60-100ms
	sMMN	Average (20 ms window)	100-200ms
	P300	Average (100 ms window)	180-400ms

tactile stimulation of individual digits is related to variation in musical experience (Elbert et al., 1995; Hirano et al., 2020). Various other activities, including video game play, could also promote greater differentiation of digits. As such, the current study explored two aspects of individual experience: a behavioral measure of finger independence and self-reported time spent playing video games on a console.

We had several hypotheses:

1. The tPCA would identify ERP components matching those reported by Shen et al. (2018a), including the N80, sMMN, and P3a components. Deviant stimuli were expected to elicit more enhanced components than standard stimuli.
2. Because prior work on the sMMN was consistent in parsing the 1st digit in a different functional category than the 3rd and 5th digits (Shen et al., 2018a), We hypothesized that deviants at the first digit will produce greater differences in the SEP than when the other digits are used as deviants. We also hypothesized that, when the 1st digit is serving as the standard, we will see the greatest differences evoked by the deviants, but that the deviant difference waves will not differ from each other.
3. Participants' ability to lift individual fingers will relate to SEP difference waves when that digit is the standard, particularly for the 3rd and 5th digits, in line with the prediction that greater dexterity and movement will be associated with stronger sensory memory traces.
4. Participants reported greater use of console video games will have a stronger functional boundary effect between the thumb and other fingers. They will display the pattern from Hypothesis 3 more than those with less video game usage.

2. Methods

2.1. Participants

Sixty undergraduate students participated in the study. They ranged in age from 18 to 43 years ($M = 21.11$, $SD = 3.65$). Most participants (34, 56.6 %) identified as white, 9 (15 %) participants identified as Black/African American, 8 participants (13.3 %) as Asian, 5 as more than one race (8.3 %), 2 were another race not otherwise reported, and 2 preferred not to say (3.3 %). Most (45) participants were not Hispanic, Latinx, or Spanish (75 %). Four (6.7 %) were Puerto Rican and 2 (3.3 %) were Mexican or Chicana. Five (8.3 %) participants identified as another Hispanic, Latinx, or Spanish origin and 4 (6.7 %) as another ethnicity/origin but did not otherwise specify. Forty-one participants (68.3 %) reported a biological sex of female. Participants' gender identities included 38 (63.3 %) women, 17 (28.3 %) men, 2 (3.3 %) non-binary, 1 gender fluid, 1 transgender man, and 1 (1.6 %) undisclosed. Of the original sample of 60, four were excluded due to insufficient EEG data.

All participants were right-handed as assessed by the Edinburgh Handedness Inventory (Oldfield, 1971), had normal-to-corrected vision, reported no history of neurological illness or abnormality, and had not had hand surgery. This study was approved by the institutional review board at [blinded]. Informed consent was obtained before participation.

2.2. Procedure

After providing informed consent, participants completed a demographics questionnaire, answered questions about musical instrument and video game experience, and completed the Edinburgh Handedness Inventory. They then engaged in a test of finger independence, followed by preparation of the EEG cap, and a passive oddball task involving stimulation of three digits of the right-hand during EEG collection. During the digit stimulation and EEG collection, participants watched footage of a wildlife documentary (Planet Earth) from a DVD on an LCD monitor screen with subtitles on and the volume muted. To mask any sound associated with the delivery of the tactile stimuli, a white noise machine was used, and participants wore foam earplugs. They

were encouraged to keep their hands in a comfortable position (either on the desk in front of them, the arms of the chair, or in their lap) and ignore the stimulation to their fingers.

2.2.1. Measures of individual differences: finger independence

To quantify finger independence, a procedure was adopted from Häger-Ross and Schieber (2000). Participants sat at a table (70 cm height) and placed their palms on the tabletop with fingers comfortably spread. An Intel RealSense D435 depth camera was mounted 138 cm from the floor, directly above the center of the table, and recorded participants' hands from above while they were instructed to lift each finger without moving the other fingers. The RealSense camera uses two internal cameras with overlapping visual fields to estimate depth. Recordings of the task were viewed in the Intel RealSense Viewer and the distance from the camera to the fingertip at the peak of the lift for each digit was recorded. Finger height was calculated by subtracting this value from the distance from the camera to the table.

2.2.2. Tactile stimulation

The procedure for delivering tactile stimulation followed the methods used by Shen et al. (2018a). Fingertip stimulation was delivered through the inflation of a membrane in a plastic casing that was attached to a finger clip (see Supplementary material – Fig. S1). The membranes were used to stimulate the palmar side of 3 digits on the right hand: D1 (thumb), D3 (middle finger), and D5 (little/pinky finger). Each tactile stimulus was triggered by a TTL pulse (10 ms duration) sent to a pneumatic stimulator unit from STIM stimulus presentation software (James Long Company). The membrane began expanding 15 ms after the TTL pulse, peaked 20 ms later, then deflated over the subsequent 80 ms. Digit stimulation comprised 6 blocks, as summarized in Table 2. The interstimulus interval was 600 ms and the within-block order of stimulus presentation was randomized. Blocks 1, 2, and 3 utilized an oddball paradigm involving 1000 trials in each block. Within each block, one digit was designated as the standard and received 80 % of the tactile stimuli while the remaining digits were designated as the deviants, and each received 10 % of the stimuli. D3 was the standard in Block 1 (replicating Shen et al., 2018a); D5 was the standard in Block 2; D1 was the standard in Block 3. Blocks 4–6 served to collect control responses, with each block consisting of 1 min of stimulation to one of the three digits. The order of Blocks 4–6 was randomized.

2.2.3. EEG acquisition

EEG signals were acquired from 32 electrodes (Fp1, Fp2, Fp3, Fp4, F7, F3, Fz, F4, F8, FC5, FC1, FC2, FC6, T7, C3, Cz, C4, T8, CP5, CP1, CP2, CP6, P3, Pz, P4, P8, POz, O1, Oz, O2, M1, and M2) secured in a stretch cap (ANT Neuro) according to the International 10–20 format. Each electrode site was filled with a small amount of conductive gel. EEG signals were collected referenced to Cz with an AFz ground and were re-referenced offline to the average of the left and right mastoids before further analysis. All EEG signals were amplified by optically isolated, high input impedance ($>1\text{ G}\Omega$) bioamplifiers from SA Instrumentation

Table 2
Example of the blocking for the tactile stimulation protocol.

Block	Digits stimulated	Outcome
1	D3 (standard)	D1 (vs. D3) deviant ERP
	D1 (deviant)	D5 (vs. D3) deviant ERP
	D5 (deviant)	
2	D5 (standard)	D1 (vs. D5) deviant ERP
	D1 (deviant)	D3 (vs. D5) deviant ERP
	D3 (deviant)	
3	D1 (standard)	D3 (vs. D1) deviant ERP
	D3 (deviant)	D5 (vs. D1) deviant ERP
	D5 (deviant)	
4	D1	D1 control ERP
5	D3	D3 control ERP
6	D5	D5 control ERP

(San Diego, CA) and were digitized using a 16-bit A/D converter at a sampling rate of 512 Hz using Snap-Master data acquisition software (HEM Data Corp.). Bioamplifier gain was 4000 and hardware filter settings were 0.1 Hz (high-pass) and 100 Hz (low-pass), with a 12-dB/octave roll-off.

2.3. Data analysis

2.3.1. Preprocessing of EEG data

Physiological data was pre-processed in MATLAB (R2022b) using a customized EEGLAB script (Delorme and Makeig, 2004) to resample the data to 512 Hz, merge the event and EEG data files and reference to the average of the mastoids. Event times were adjusted by 20 ms to account for the delay between the trigger, the solenoid opening, and the pulse reaching the stimulator. Data files were visually inspected to ensure signal quality for both mastoid electrodes before re-referencing. Files in which either one or both mastoid electrodes were problematic were not included in further analyses ($n = 1$).

To avoid human error during visual artifact deletion, the data were cleaned using the HAPPE+ER v3.1 low-density pipeline (Gabard-Durham et al., 2018; Lopez et al., 2022; Monachino et al., 2022). The HAPPE+ER pipeline employs third-party software functions, including EEGLAB (Delorme and Makeig, 2004), Cleanline (Mullen, 2012), and FASTER (Nolan et al., 2010). Artifacts were detected by wavelet thresholding using HAPPE+ER's "soft" threshold and bad channels were identified using HAPPE's default specifications. We used two FIR filters at 30 Hz low-pass, 0.1 Hz high-pass, and a 60 Hz line noise within the Cleanline program. Data were separated into epochs of 100 ms before the event and 500 ms after (−100 to 500 ms), and were baseline corrected from −100 to 0 ms. Epochs were rejected based on amplitude (−150 to 150 μ V) and joint probability; they were later interpolated using the FASTER program after which point, the previously identified bad channels were interpolated. The HAPPE+ER pipeline provides quality assessments and output that can be used for inclusion criteria. We used three variables for inclusion criteria: high correlations between the data pre and post line noise removal (above or equal to 0.90 at 59 and 61 Hz), >40 % usable epochs per condition, and >70 % usable channels (HAPPE+ER User Guide, Monachino et al., 2022). Three participants were excluded based on these criteria and were not included in the final analysed sample size of 56.

A series of regressions with Tukey adjusted pairwise comparisons were run to test for differences in the number of usable trials across locations. There were no differences among the standards (all p 's > 0.99) or the controls (all p 's > 0.14). Within the deviant pairings, the D3 deviant to D1 standard had slightly fewer trials ($M = 66.6$, $SE = 0.93$) than D1 deviant to D5 standard ($M = 69.7$, $SE = 0.93$), $t(255) = -2.89$, $p = .048$.

2.3.2. Identification and scoring of ERP components

After running the epochs through HAPPE+ER, two temporal principal component analyses (tPCAs) based on Scharf et al. (2022) were conducted on the data using R Statistical Software (v4.3.0; R Core Team, 2021) and RStudio ("Cherry Blossom" Release) (RStudio Team, 2020) to identify and score separate SEP components. The resulting 'parts' of a tPCA are usually called components, but here we will refer to them as factors so they cannot be confused with SEP components. We opted for a restricted solution, where the number of factors to extract was much smaller than the number of sampling points (which is used in an unrestricted solution). A parallel analysis (Horn, 1965) and an Empirical Kaiser Criterion (Braeken and van Assen, 2017) identified the factors to be extracted. We then performed an oblique rotation because it allowed for correlations between factors and, due to the low spatial resolution of EEG, the factors resulting from a tPCA will be correlated (Dien and Frishkoff, 2005; Scharf et al., 2022; Scharf and Nestler, 2018). As recommended by Scharf et al. (2022), a Geomin (Yates, 1987) rotation (Bernaards and Jennrich, 2005) with 30 random start values was used

and a rotation epsilon parameter of 0.01. The first tPCA was conducted to identify the early negative component (N80) and included all deviant and control conditions.

2.3.2.1. Identity MMN. The second tPCA focused on the sMMN and the P300, both of which are calculated by difference scores (Zheng et al., 2012). Instead of subtracting the standard from the deviant to compute a difference waveform, here we used the identity MMN method, which involves stimulating each location in an uninterrupted train during "control" blocks that are separate from blocks in which the oddball paradigm was presented (Shen et al., 2018a). The difference waveform was then computed by subtracting the ERP elicited to stimulation of one location during the control blocks from the ERP elicited when the same location was the location of the deviant stimulus during the oddball paradigm (Fu and Monahan, 2021). This approach allowed the researchers to control for any differences in tactile sensitivity between the different fingers, replicating the methods from the prior study on which the current work was based (Shen et al., 2018a). The second tPCA used only the difference conditions. Waveform graphs were created with ggplot2 (Wickham, 2016), while topography plots were created using the eegUtils package (Craddock, 2022). Factors and components were matched based on factor topography, waveform, and peak latency.

2.3.3. Hypothesis testing and data analysis plan

2.3.3.1. Digit categories vs. individualization. We performed separate random intercept model analyses for each SEP component using R Statistical Software (v4.3.0) (R Core Team, 2021) and RStudio (RStudio Team, 2020; "Cherry Blossom" Release). The data were imported, organized, and inspected with various packages, including readr (Wickham et al., 2023b), psych (Revelle, 2023), and dplyr (Wickham et al., 2023a). Outliers present in the data were identified as values more extreme than the confidence interval of the median ± 3 *median absolute value (Leys et al., 2013). Outliers were winsorized using the datawizard (Patil et al., 2022) package, using the most extreme values inside of the confidence intervals as the threshold points. The control variables included were age, gender, race, and ethnicity. We also added condition, region, and hemisphere as focal variables, where applicable, within the multilevel regressions. All regressions were conducted using lmerTest (Kuznetsova et al., 2017).

Based on previous work (Shen et al., 2018a) the region of interest for the N80 comprised electrodes C3, C4, F3, F4, FC1, FC2, FC5, and FC6. Variables included condition (D1 control, D5 control, D3 control, D1 deviant (D3 standard), D5 deviant (D3 standard), D3 deviant (D5 standard), D1 deviant (D5 standard), D3 deviant (D1 standard), D5 deviant (D1 standard)), region (frontal/frontocentral/central), and hemisphere (left/right).

For sMMN, the region of interest was electrodes C3 and C4 because this is the location of the somatosensory cortex and mean waveforms from prior work demonstrated a differential effect of digit stimulation here (Shen et al., 2017). The conditions were D1 difference (D5 standard), D1 difference (D3 standard), D5 difference (D3 standard), D5 difference (D1 standard), D3 difference (D5 standard), and D3 difference (D1 standard). Two separate multilevel regressions were conducted for each SEP component, without interactions and with one interaction (condition*hemisphere) to examine unconditional main effects.

For P3a and P3b, the region of interest was restricted to Cz because prior research found the strongest effects for this site (Shen et al., 2018a). As such, the multilevel regression only included control variables and the condition variable.

To probe the main effects, Tukey-adjusted pairwise t -tests were performed to assess the directionality of effects of factors with more than two groups. Simple slope and simple effect analyses were conducted on significant main effects and interactions via the emmeans (Lenth, 2023) package to understand the directionality of the relationship between the

variables. Graphs were created using the effects (Fox and Weisberg, 2018a, 2018b), ggplot2 (Wickham, 2016), performance (Lüdtke et al., 2021), and palettee (Hvitfeldt, 2021) packages.

2.3.3.2. Exploratory analyses: individual differences and experience. To explore the relation of finger independence and the SEP waveforms, we ran regressions involving the sMMN, P3a, and P3b. Individual digit height (our metric of finger independence) was assessed as a predictor when that digit was used as the standard (e.g., one regression predicting the influence of D5 height when D5 is the standard). The regressions controlled for the same variables as the main regressions described in the previous section.

We examined the impact of overall video game experience on the sMMN by rerunning the original two regressions and limiting the scope to only include deviants when D1 was the standard. This is because console-based video games rely heavily on the thumbs to control movement and actions. Therefore, we hypothesized this experience would be highly localized. Initially, our predictor was a dichotomous (yes/no) video game experience variable, and then using only the subset of the participants with video game experience, the regression was redone using a continuous variable of the amount of video game hours per week instead of the dichotomous experience variable.

3. Results

3.1. Principal component analyses

We examined the topography and timing of each factor identified by the tPCAs to find the best match with our components of interest. There was no cut-off based on the amount of variance explained by an individual factor. Scharf et al. (2022) explain that there usually are around 8–12 major factors extracted, while the rest of the factors are minor factors and usually represent noise or latency jitter between participants within the data (see also Dien, 2012). Visual inspection of factor graphs ensures that chosen factors are not simply noise, as minor factors look peculiar. Supplementary Fig. 2 contains all grand mean waveforms.

In the first tPCA, we included all deviant and control conditions, which led to 21 factors being identified. There were 8400 cases (multiplying participants, recording locations, and conditions) and 308 variables (sampling points) for this tPCA. There was one factor (number 12; amount of variance explained was 2 %) that had a latency of 86 ms, which is around 80 ms like the ERP component N80 and was characterized by a more negative frontal region with a more positive parietal region. Given this latency and scalp distribution, we took this factor as being the N80. In the grand mean waveforms, the N80 component manifested as a positive peak in the grand mean waveform with a peak latency at 86 ms. However, the topographical plot localizes positive amplitude in the central/temporal and negative activity in the frontal region.

The second tPCA was run on the difference waveforms and included all digit comparison conditions. This also led to 21 factors being identified. There were 10,080 cases (multiplying participants, recording locations, and conditions) and 308 variables (sampling points) for this tPCA. One factor (factor 9; amount of variance explained: 2 %) matched to the sMMN component through a peak latency of 130 ms and maximal amplitude in the difference waveform over electrode C3. For a longer latency ERP component, all factors after 200 ms were positive, which indicated the presence of a P300. However, there was not one singular component around 200–300 ms that localized around electrode Cz, where the P300 would be expected. There were two factors, one with a peak latency at 215 ms (factor 5, amount of variance explained: 5 %) with a strong Cz localization and one at 260 ms (factor 1, amount of variance explained: 37 %) with a slightly more frontally localized topography. Based on the latency and scalp distributions, we concluded that these factors are ERP components P3a and P3b, respectively. There

was one additional factor (factor 4) with a peak latency at 354 ms; however, this was ruled out because its topographical distribution was too widespread, spanning CP1, P3, and Pz. Topography was prioritized because ERP component timings are known to change based on modality. Component waveforms (vs. the grand mean waveforms) and their topographies are presented for all factors in Fig. 1.

3.2. N80, sMMN, and P300

The unconditional random intercept model for component N80 showed significant main effects of hemisphere, region, and condition. N80 was significantly enhanced (more negative) over the right hemisphere than the left ($\beta = 1.37$, SE = 0.08, $t(3621) = -17.25$, $p < .001$) and over the frontal region as compared to both central ($\Delta\beta = -2.33$, SE = 0.11, $t(3621) = -20.72$, $p < .001$) and frontocentral regions ($\Delta\beta = -0.71$, SE = 0.10, $t(3621) = -7.32$, $p < .001$). Across task conditions, N80 amplitudes were significantly enhanced for the D3 control condition compared to the D5 control ($\Delta\beta = -0.53$, SE = 0.17, $t(3621) = -3.16$, $p = .04$), D1 deviant to D3 standard ($\Delta\beta = -0.58$, SE = 0.17, $t(3621) = -3.45$, $p = .02$), and to the D5 deviant to D1 standard ($\Delta\beta = -0.52$, SE = 0.17, $t(3621) = -3.07$, $p = .05$) (see Fig. 2A). The conditional model indicated a significant interaction between hemisphere and condition. This interaction was driven by more negative N80 amplitudes over the left hemisphere compared to the right hemisphere (all $t(3621) < -4.79$, all p 's < 0.002).

The unconditional random intercept model for the sMMN found significant main effects of hemisphere and condition. The sMMN amplitude was more negative over the left hemisphere compared with the right ($\beta = -1.10$, SE = 0.13, $t(561) = -8.75$, $p < .001$). There was also an effect of condition (visualized in Fig. 2B) that was driven by a larger (more negative) sMMN response to the D3 difference (D5 standard) condition compared to D5 difference (D3 standard) ($\Delta\beta = -0.66$, SE = 0.22, $t(561) = -3.01$, $p = .033$). The conditional model did not yield any significant interaction results, all p 's > 0.15.

The P3a analysis revealed a significant effect of condition (see Fig. 2C). Follow up analyses showed that the D3 difference (D1 standard) condition had a lower amplitude P3a than the D1 difference (D3 standard) ($\Delta\beta = -1.66$, SE = 0.34, $t(255) = -4.86$, $p < .001$), D1 difference (D5 standard) ($\Delta\beta = -1.27$, SE = 0.34, $t(255) = -3.70$, $p = .004$), and D3 difference (D5 standard) ($\Delta\beta = 1.22$, SE = 0.34, $t(255) = 3.55$, $p = .006$) conditions.

For the P3b, there were no significant differences across conditions, all p 's > 0.11. P3b amplitudes by condition are shown in Fig. 2D. Supplementary Fig. 3 also shows component waveforms across each condition.

3.3. Individual differences

3.3.1. Finger independence

Three participants were missing data on finger independence. The remaining participants averaged lifting distances as follows: D1 21.4 mm (SD = 8.4), D2 21.8 mm (SD = 7.3), D3 22.2 mm (SD = 7.3), D4 18.8 mm (SD = 6.2) and D5 18.1 mm (SD = 5.8). There were no significant main effects or interactions in the regression involving finger independence and sMMN or P3b amplitudes. For the P3a, there was a significant interaction between D5 height and condition (displayed in Fig. 3). While in both conditions D5 served as the control, greater digit mobility was associated with enhanced P3a amplitude compared to when D3 was the deviant, which showed the inverse relationship ($\beta = 0.81$, SE = 0.30, $t(48) = 2.76$, $p = .008$).

3.3.2. Video game experience

Twenty-seven (46 %) of our participants reported regular experience using a video gaming console. Weekly game time ranged from 0.5 to 15 h ($M = 5.21$, $SD = 4.43$). In the random intercept model assessing gaming experience as a dichotomous variable (Y/N), there was a significant

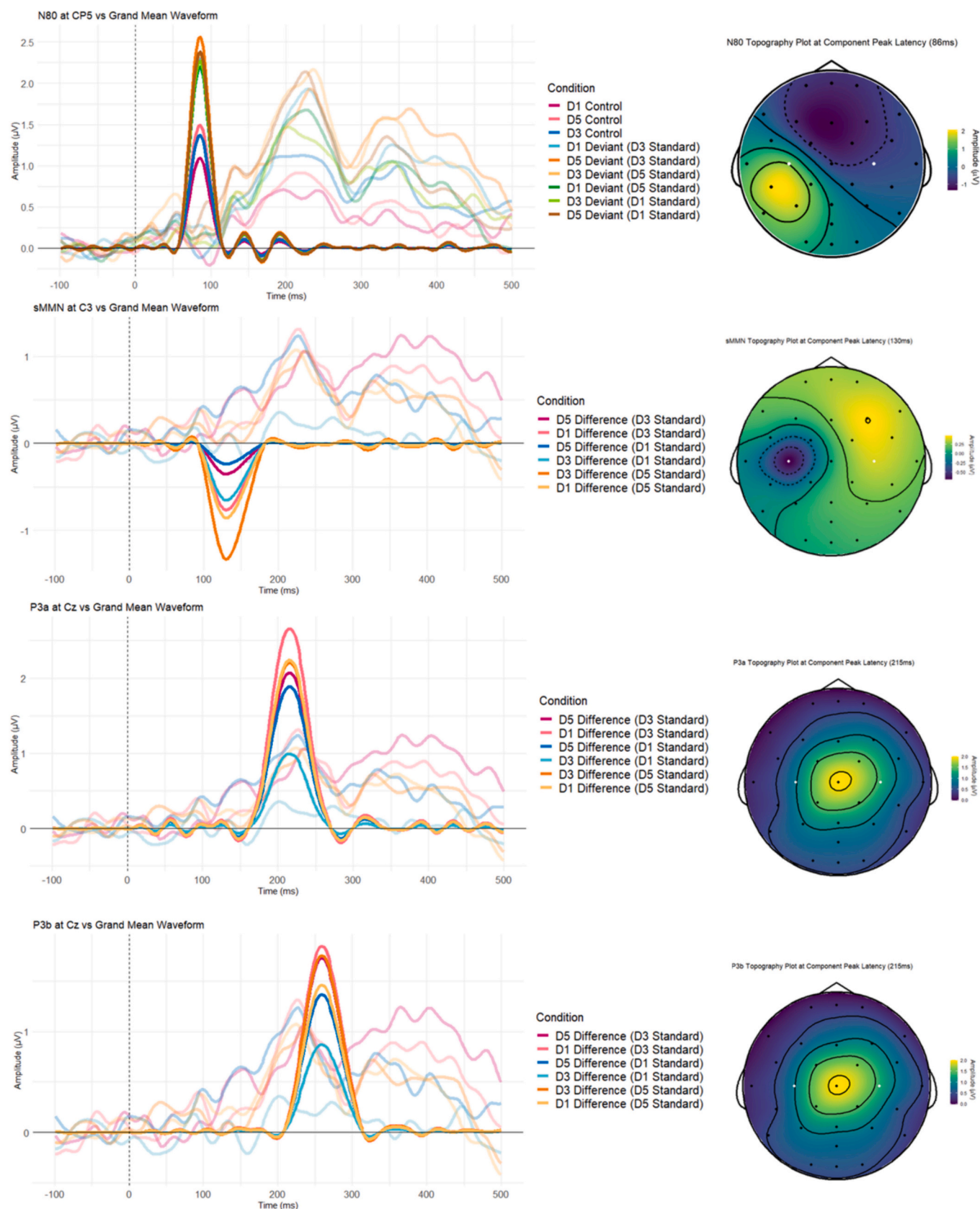


Fig. 1. ERP component (bold lines) and grand mean (faded lines) waveforms are displayed in the left panel. Component topographies at peak latency are displayed in the right panel.

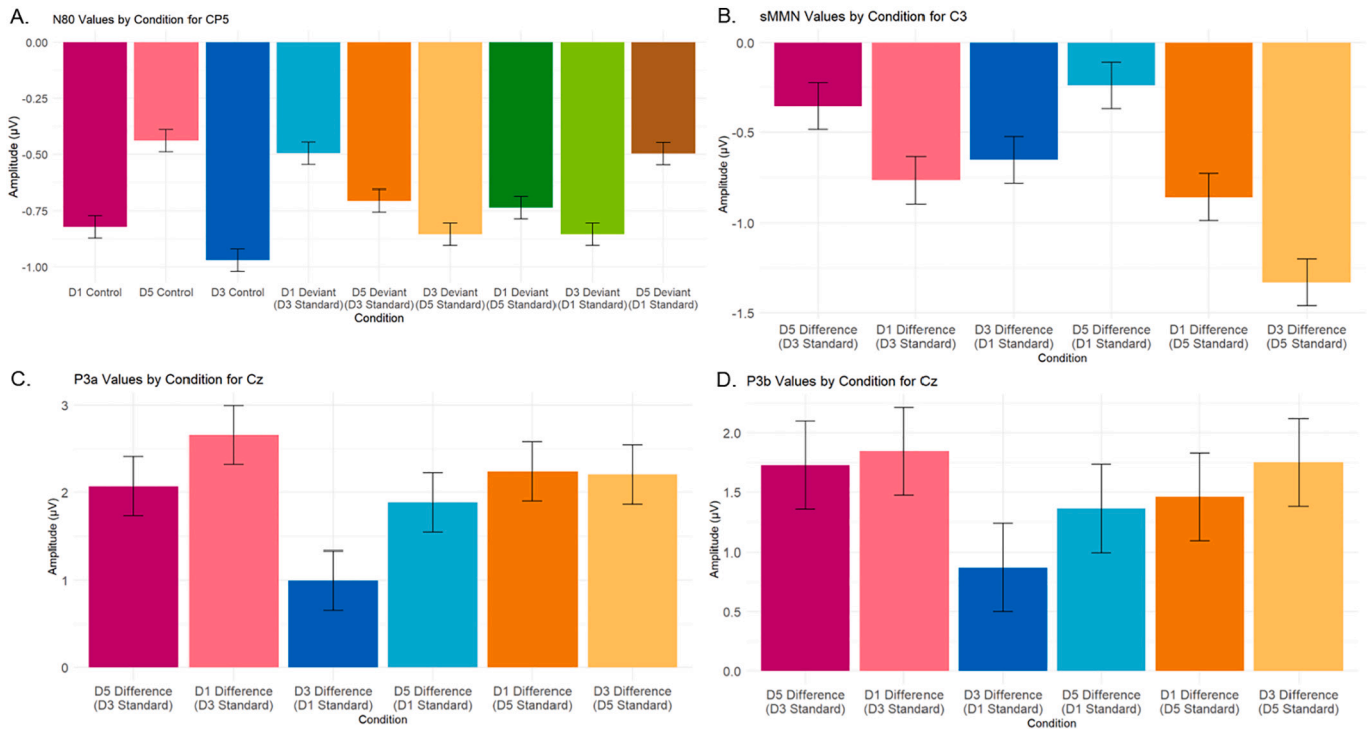


Fig. 2. Average component amplitudes by condition at most prominent channel. For all except N80, values reflect the identity MMN method of deviant minus control, where the control matches the standard site.

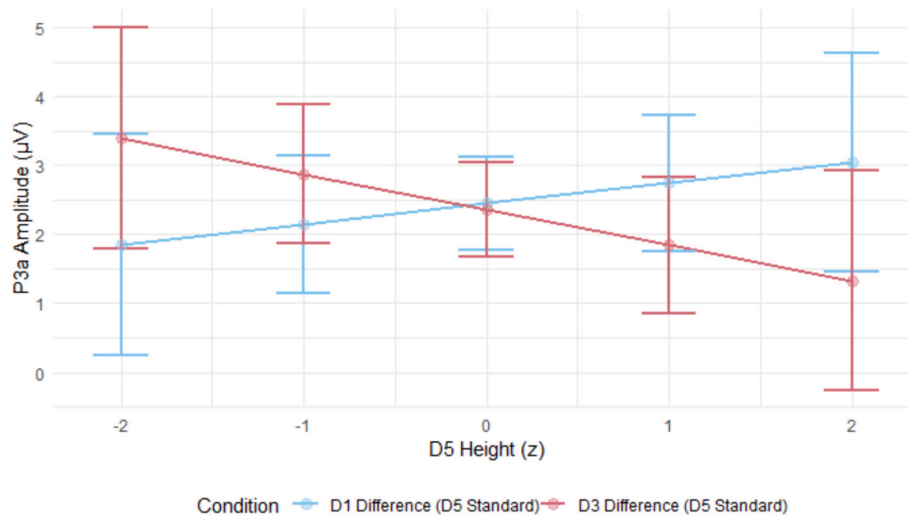


Fig. 3. P3a amplitude by D5 mobility and condition, capturing the inverse distribution based on condition and digit mobility.

main effect of experience, whereby participants without video game experience had an enhanced sMMN component ($\beta = 0.81$, $SE = 0.36$, $t(51) = 2.28$, $p = .027$). There was also a trending main effect of condition, where D5 difference (D1 standard) had an enhanced sMMN compared to D3 difference (D1 standard) ($\Delta\beta = 0.38$, $SE = 0.20$, $t(153) = 1.93$, $p = .056$). The conditional model showed a significant interaction between video game experience and condition, driven by the overall difference in amplitude between those with and without video game experience. Within those without video game experience, sMMN for the D3 difference (D1 standard) condition was significantly enhanced compared to the D5 difference (D1 standard) condition ($\Delta\beta = -0.71$, $SE = 0.27$, $t(153) = -2.67$, $p = .042$), as shown in Fig. 4. Interestingly, sMMN amplitude was greater in those who did not play video games, but the differential impact of condition disappeared in

those that did. There was no impact of the continuous measure of video game experience on sMMN amplitude.

4. Discussion

The present study sought to replicate and extend previous research on SEPs evoked during passive tactile oddball paradigms involving stimulation of different body parts. One basis for the current study was the work of Shen et al. (2018a) who employed a tactile oddball paradigm in which the middle finger (3rd digit) served as the location of the frequent standard stimulus and the thumb (1st digit), and little finger (5th digit) served as the locations of infrequent deviant stimuli. In an extension of this work, the current study used the same mechanical stimuli in a more comprehensive oddball paradigm in which each of the

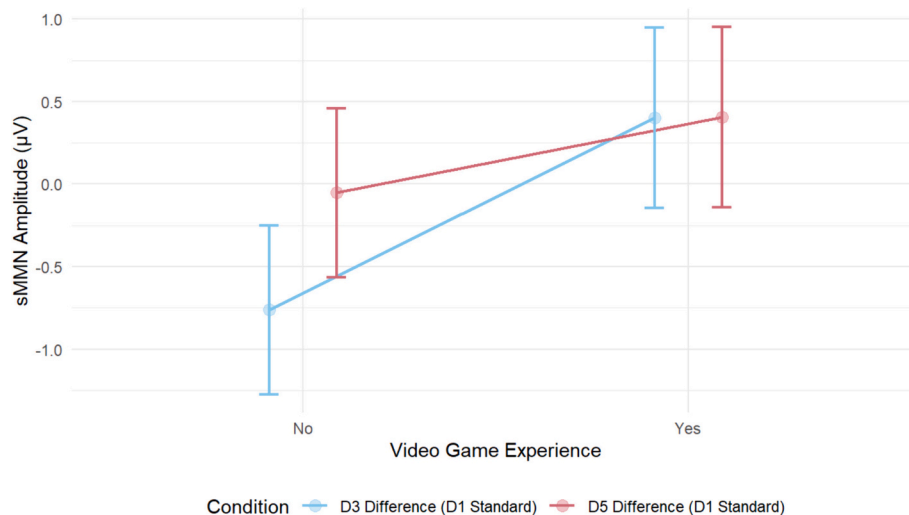


Fig. 4. sMMN amplitude by video game experience, split dichotomously, and condition, highlighting the interaction between each variable.

3 digits receiving stimulation (1st, 3rd, and 5th) served as both standard and deviant locations. Further, the SEP components from the tactile oddball paradigm were examined in relation to individual differences in the ability to independently move each finger, and to variation in the extent of time spent playing video games on a regular basis. Our interest in the individual difference analyses was driven by the suggestion of Shen et al. (2018a) that greater finger individuation may be associated with larger sMMN responses across different fingers.

We employed a data-driven approach to the analysis of the SEP response, mapping factors derived from a PCA onto traditional SEP components, specifically the N80, sMMN, and P300. The N80 component waveform showed a particular scalp distribution with negative values across the frontal region and more positive values at left parietal sites. In line with the results of Shen et al. (2018a), the sMMN was present between 100 and 200 ms and was strongest (most negative) over the left central region, contralateral to the tactile stimulation. With regard to later components, the PCA indicated the presence of both the P3a and the P3b. Teasing apart these subcomponents is often only possible using a PCA (Polich, 2007), which may explain why they were not separated in past research.

The appearance of P3b is somewhat surprising because participants were engaged in a passive oddball paradigm. Typically, the P3b is elicited only during active oddball paradigms in which participants must act on the deviant stimuli (e.g., by pressing a button when a change in the stimulation train is perceived). One potential explanation of the appearance of the P3b in a passive task paradigm is that participants allocated attentional resources to the tactile deviants, even though the task did not call for these stimuli to be actively attended to (Polich, 2007). Additionally, the P3b appears earlier, at 260 ms following stimulus onset, than may be expected. Prior work on which the present study was based utilized a window of 180–400 ms to capture the P300 response (Shen et al., 2018a) and research using a tPCA in an auditory oddball paradigm captured P3b from 255 to 350 ms (Barry et al., 2020). More research is needed with analytical techniques capable of parsing the P3a and P3b across oddball paradigms.

Amplitudes of each SEP component were compared across the various digit pairings. Given that the N80 is considered part of an early change detection mechanism, it was surprising that one of control conditions (D3) produced the largest N80 response. Visual inspection of the results suggests that the pattern of sMMN responses (Fig. 2B) replicated prior results in that sMMN amplitude was larger when D3 was the standard and D1 was the deviant than when D5 was the deviant (Shen et al., 2018a). However, further complexity is suggested by the finding that there was a significant difference between sMMN amplitude for the

D3 difference (D5 standard) and D5 difference (D3 standard). This complicates the interpretation of the findings of Shen et al. (2018a) in terms of the functional relations between different digits. The pattern of P3a and P3b responses across conditions were similar to each other. We replicated the pattern of past findings in the D3 standard comparisons (Shen et al., 2018a). However, the pattern was reversed compared to the sMMN for the D1 standard comparisons; D5 produced a more robust response than D3. Where P3a and P3b diverge is in the amplitude of difference between the D3 and D5 standard comparisons: This difference is lessened from P3a to P3b when D3 is the standard and increased when D5 is the standard. Taken together, these results do not support our second hypothesis and suggest that when the full range of digit combinations is tested, that more complexity arises beyond prior results (Shen et al., 2018a).

Further analyses examined the relations among SEP amplitudes and individual differences in finger independence and video game experience. When accounting for individual differences in finger independence, we found that only the ability to isolate D5 movement was related to P3a amplitude. The D1 deviant comparison followed our hypothesized direction, with greater D5 mobility predicting a more robust P3a amplitude. However, the D3 deviant produced an inverse pattern. We hypothesized more mobility would equate to greater finger differentiation, leading to stronger standard stimuli memory traces and a more robust SEP response to the deviant.

Contrary to our hypothesis, participants who did not play video games had a more negative sMMN indicating they formed a more robust sensory memory trace, particularly between D1 and D3. Interestingly, among those who did play video games, there was no difference between digits that were the location of the deviant stimuli. This effect did not persist when video game experience was entered as a continuous variable. As such, it does not seem that this motor behavior is salient enough to promote increased functional differentiation of digits as indexed by SEP measures. An alternative explanation for the lack of findings related to individual differences is that we relied on self-reported time spent playing video games, and we did not directly assess gaming proficiency. The inclusion of more diverse measures of experience with activities that might promote finger differentiation may be helpful for future work.

Other manual experiences, such as musical experience, have also been found to impact ERP responses (Elbert et al., 1995; Hirano et al., 2020). In the present study, we attempted to collect data on musical experience but were unsuccessful in accruing a sufficient sample to be included in the formal analysis. The literature suggests that it is important to control for the type of instrument played and experience levels and our convenience sample spanned a variety of both, limiting

statistical power. While lesser studied, technology use, including computers, phones, and tablets, could also be relevant within the present study. We attempted to capture some of this with a measure of typing speed, but it was not significant as a predictor and was unrelated to a specific, uniform strategy (e.g., not all fast typers used the home key method). Similarly, although unstudied, tablet use involves a diverse set of strategies such as stylus pen, single finger, or finger combinations, which could also be applied to phone use. As our sample comprised college students who tend to have a uniformly high smartphone usage, we did not anticipate meaningful individual differences.

Additional research is needed to further elaborate on our results and address study limitations, including additional counterbalancing and a more conservative approach to identifying the sMMN component. Regarding the former, the order of the standards as well as the deviants should be counterbalanced to control for possible order effects. While we replicated the method of prior research and relied on latency and topography to identify the sMMN, one potential method would be a polarity inversion check after re-referencing to the nose (Duncan et al., 2009). Future work could consider using an active oddball paradigm to test whether the P3b effects are similar to the current findings, or whether they become more distinct. However, it should also be noted that adding attentional or memory requirements may also change the neural representations of digits, as suggested by a recent study (Rabe et al., 2023).

In conclusion, the current study adds to the literature by including a fully matched tactile oddball paradigm design in which every location of the standard stimulus was also used as a location of deviant stimulation. We replicated findings from prior research using a tactile oddball (Shen et al., 2018a) and extended this work to shed light on patterns of individual differences. Our results also mark one of the first instances in which a data driven PCA approach was used to identify ERP components elicited to tactile stimulation.

CRedit authorship contribution statement

Melissa N. Horger: Writing – review & editing, Writing – original draft, Project administration, Methodology, Investigation, Formal analysis, Data curation. **Kaitlyn Campbell:** Writing – review & editing, Writing – original draft, Visualization, Formal analysis. **Peter J. Marshall:** Writing – review & editing, Writing – original draft, Supervision, Methodology, Conceptualization.

Declaration of Generative AI and AI-assisted technologies in the writing process

No part of this manuscript was written with generative AI or AI-assisted technologies.

Declaration of competing interest

Authors have no conflicts of interest to disclose.

Data availability

Data will be made available on request.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.ijpsycho.2024.112391>.

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